TutorConnect Milestone 4 report

Enterprise AI functionalities

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1 AI integration and application design

TutorConnect now has a conversational scheduling assistant, an autonomous reminder agent and policy answers grounded in a small knowledge source. Both agents use Spring AI with the Gemini Developer API. The existing session login, JDBC repositories and transaction services remain responsible for appointment rules.

The conversational agent can query available sessions, list the customer's appointments and propose a booking or cancellation. A separate confirmation request performs the write. The reminder agent runs on a schedule, asks Gemini to generate a study reminder and sends the result through the notification service built in Milestone 3.

Spring AI 2.0.2 is pinned alongside Spring Boot 4.1.1 and Java 17. AiModelConfiguration creates the Google GenAI client with an external API key and vertexAI(false). No project ID or location is configured. The chat model defaults to gemini-3.8-flash; the text embedding model is gemini-embedding-001. Environment variables allow either model to be changed when account access or the model catalog changes.

2 Conversational agent and tools

AiChatService uses Spring AI ChatClient and a ToolCallingAdvisor. The model can call a tool, read its result and call another tool before answering. For example, it can query the next seven days, select an available algebra session and propose that session. An unavailable slot produces a conflict result that tells the assistant to query again and offer alternatives.

Tool	Application behavior
checkAvailability	Calls SlotService for a UTC date range of at most 31 days and returns up to 50 sessions, earliest first
listMyAppointments	Calls AppointmentService.customer for the authenticated customer's ID
book	Reads the real slot and creates a booking proposal with a server-generated summary
cancel	Checks ownership and future BOOKED status before creating a cancellation proposal
searchKnowledge	Retrieves policy passages from SimpleVectorStore, including source IDs

Each request creates an AiBookingTools instance bound to the authenticated Account and its server-side conversation. Tool schemas contain appointment and slot parameters but no user ID, role or confirmation flag. The model cannot choose the account whose appointments it accesses.

```
var tools = new AiBookingTools(
    actor, state, slots, appointments, knowledge);
// ChatClient executes the registered tools in its advisor loop.
client.prompt().messages(history).tools(tools).call();
```

The HTTP session keeps up to 20 user and assistant messages for follow-up questions. It accepts no client-supplied history. A turn allows at most eight model rounds and ten tool executions. The assistant is instructed to clarify ambiguous dates or subjects and answer in the user's language. These language behaviors depend on the model; the server enforces identity and write protection independently.

3 Confirmation and the transactional write path

A booking tool call does not insert an appointment. It produces a pending action containing the exact slot, service, summary and a random token. The pending action belongs to one customer session, expires after five minutes and is replaced when a new chat turn begins. Only the browser receives its token; the tool result sent back to Gemini contains the summary and confirmation requirement.

The customer reviews the summary and clicks Confirm this action. That sends a CSRF-protected POST to /api/customer/ai/confirm. The endpoint reads the stored action and compares the token, identity and expiry. It accepts no replacement slot or service fields. Saying yes in chat cannot authorize a write.

```
var pending = state.pending;
// Check actor, token and expiry against server-owned state.
state.pending = null; // consume before attempting the write
long id = appointments.book(
    actor.id(), pending.targetId(), pending.serviceId());
```

The confirmed booking calls the existing AppointmentService and its separate BookingTransaction bean. The READ_COMMITTED transaction locks the availability row with SELECT FOR UPDATE, rechecks availability and inserts the appointment. The generated active_slot_id unique key remains the database backstop. A second customer who confirms the same slot receives a 409 conflict after the first transaction wins. The appointment notification is still queued atomically and sent after commit.

Cancellation follows the same confirmation mechanism and calls AppointmentService.cancel. Ownership and future BOOKED status are checked again in the transactional service. Consuming the token prevents duplicate writes after a repeated request. If a conflict occurs or the proposal expires, the customer must request a new proposal. Session state and confirmations are synchronized within the current application instance.

Both customer routes retain CUSTOMER role checks and browser CSRF protection. A provider cannot use the customer assistant, and another login session cannot confirm a copied token. Clear conversation and Discard also remove pending actions.

4 Autonomous reminder agent

AiReminderService has a genuine @Scheduled job with a default delay of 60 seconds. It scans up to 20 BOOKED appointments starting within the next 24 hours and excludes appointments that already have a reminder event. The schedule runs without a chat prompt. Providers can also run it from the dashboard through POST /api/provider/ai/reminders/run; that request scans only the logged-in provider's appointments.

For each candidate, ChatClient asks Gemini to write a short reminder using the customer's name, tutoring subject and UTC start time. It requests a preparation suggestion and rejects blank or oversized output. A model failure is logged and leaves the appointment eligible for another cycle. The code does not substitute fixed text and present it as an AI-generated reminder.

Generation happens before the database transaction. ReminderTransaction then locks and rechecks the appointment, including its BOOKED status and future time, before inserting an APPOINTMENT_REMINDER event. This prevents a cancellation during generation from producing a new reminder.

```
String text = client.prompt()
    .system(reminderInstructions).user(appointmentFacts)
    .call().content();
if (transaction.enqueue(appointmentId, text)) {
    dispatcher.afterCommit(appointmentId, "APPOINTMENT_REMINDER");
}
```

The event ID is appointment:<id>:APPOINTMENT_REMINDER. The outbox primary key and appointment lock prevent duplicate reminder records. NotificationDispatcher uses the existing HTTP gateway with bounded retries and persistent fallback. A failed delivery remains PENDING for recovery. A reminder that was cancelled or has already started before recovery is marked FAILED with NOT_UPCOMING and is suppressed. The receiver's durable event ID deduplication protects repeated HTTP attempts.

The outbox makes delivery durable after generation and enqueue; it does not persist an unfinished LLM request. HTTP delivery is at least once, and the mock stores one receipt per event. It sends no real email or SMS. A cancellation can still race an HTTP request already in flight, so suppression cannot retract a notification already accepted by the receiver.

Scheduling is enabled independently of notification recovery. A two-thread scheduler allows outbox recovery to run while the reminder job waits for a model. The GenAI client has a 20-second request timeout and Spring AI permits one retry. Logs record event or appointment IDs and failure categories without logging keys or full model responses.

5 Knowledge source and grounded answers

The knowledge file knowledge/tutorconnect-policies.md contains six short, hand-written passages about cancellation, available hours, services, booking, reminders and access rules. Its cancellation policy matches the application: customers may cancel their own future BOOKED appointments, while past or cancelled appointments cannot be cancelled again. The file defines no refund policy.

AiKnowledgeService splits the file at its policy headings. Each passage becomes a Spring AI Document whose ID and source metadata identify the policy. On the first search, the embedding model creates real Gemini vectors and SimpleVectorStore stores them in memory. A successful index is reused for the life of the JVM; a failed ingestion is retried on a later request. Restarting the app rebuilds the index when it is next needed.

```
var store = SimpleVectorStore.builder(embeddingModel).build();
store.add(policyDocuments);
var hits = store.similaritySearch(SearchRequest.builder()
    .query(question).topK(3).similarityThreshold(0.35).build());
```

The searchKnowledge tool returns up to three matching passages with source IDs such as policy-cancellation. The assistant receives those passages as reference data and is instructed to cite them, answer only supported policy facts and admit when the source lacks an answer. Live availability and personal appointment questions use their own service tools, because the FAQ cannot establish current database state.

Retrieval grounds the prompt but does not guarantee that every generated sentence is correct. The knowledge file is a small demo policy set. Neither the vector store nor the model is allowed to bypass the server's confirmation and authorization checks.

6 Validation and configuration

On November 11, 2026, the full suite passed on Java 17 with both H2 and MySQL 8.4. Each run had 81 tests, zero failures, zero errors and zero skipped tests. The Java 17 application image also built successfully. The source retains the GitHub Actions workflow that builds and tests on push; hosted results are separate from these local runs.

Test group	Count	Evidence
Existing Milestone 2 and 3 suites	52	Booking, role checks, locking, rollback and notification recovery
AiWorkflowTests	13	Real Spring AI tool loop, proposals, explicit confirmation, two-customer race, ownership, CSRF, RAG and reminder trigger
AiFailureTests and AiDisabledTests	4	Safe model errors, bounded loops, no fabricated reminders and app availability without a key
AiConfirmationTests	4	Expiry, identity, role checks and bounded session history
AiReminderDeliveryTests	4	HTTP delivery, one event per appointment, cancellation rechecks and provider scope
AiSchedulingTests	2	Scheduling independent of notification recovery and two scheduler threads
GeminiAdapterTests	2	Actual Google SDK with local HTTP responses, signed tool round trip and filtering thought parts

The concurrency test uses two independently authenticated HTTP clients. Both obtain a proposal for the same slot, wait at readiness/start latches and confirm through futures. It asserts one 200 response, one 409 response and exactly one active database appointment. The winning write also follows the existing notification outbox path.

GeminiAdapterTests found and reproduced a Spring AI 2.0.2 response issue: thought summaries can appear before public answer parts. GeminiPublicAnswerModel removes those summaries, preserves tool-call signature metadata and combines public answer parts. The regression test checks both the complete answer and a signed function-call exchange.

External model responses are controlled in repeatable tests. The actual Google SDK adapter is tested against a local HTTP server, which verifies the integration format without contacting Google. Live Gemini chat, embedding quality and personalized reminder wording still require verification with the student's configured API key. These local results do not claim a successful live Google call.

AI_ENABLED enables the model configuration and GEMINI_API_KEY supplies the Google AI Studio key. The key is read at runtime; .env is excluded from Git and the Docker build context. With AI disabled, the original appointment app remains usable and AI operations return 503 with a configuration message. The run guide includes live verification commands. The student-recorded walkthrough must explain the core implementation for at least five minutes.